

IMRAN AHSAN

DATA SCIENCE • GRAPH MACHINE LEARNING • TRUSTWORTHY AI

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WEBSITE

SCHOLAR

LINKEDIN

GITHUB

ACADEMIC PROFILE

AI and data-science researcher specializing in trustworthy graph learning, with six years of university teaching and more than four years of professional software-development experience. Doctoral research focuses on efficient graph unlearning, explainability-based verification, privacy diagnostics, and realistic deletion benchmarks for graph neural networks. Broader work spans reproducible evaluation, data quality and governance, information retrieval, retrieval-augmented recommender systems, and AI software engineering. Combines method development with open research artifacts, rigorous benchmarks, and implementation-oriented teaching.

RESEARCH AREAS

Graph Neural Networks

Graph Unlearning

Explainable AI (XAI)

Trustworthy / Responsible AI

Privacy-Preserving AI

EDUCATION

Ph.D. in Engineering — Chung-Ang University, Seoul, Republic of Korea

2023–2026

Department of Smart Cities · Degree expected August 21, 2026

- Doctoral defense completed June 2026; CGPA 4.26/4.50.
- Dissertation: “Toward Efficient and Effective Graph Unlearning: Challenges, Methods, and Evaluation.”
- Supervisor: Professor Mucheol Kim

M.S. in Software Engineering — National University of Sciences and Technology (NUST), Pakistan

2014–2017

- Thesis: “Automated Test Case Generation from User Specifications using Natural Language Processing.”
- CGPA 3.25/4.00.
- Supervisor: Dr. Wasi Haider Butt

B.S. in Software Engineering — International Islamic University Islamabad (IIUI), Pakistan

2009–2013

ACADEMIC & RESEARCH APPOINTMENTS

Doctoral Researcher — Data Intelligence Laboratory, Chung-Ang University

2023–2026

- Develops methods, benchmarks, and verification protocols for graph unlearning, explainable AI, and privacy-aware model updates.
- Builds reproducible Python, PyTorch, and PyTorch Geometric pipelines covering GCN, GAT, GraphSAGE, membership-inference diagnostics, and multi-view evaluation.

Lecturer in Computer Science — National University of Modern Languages, Islamabad, Pakistan

2017–2023

- Taught Java OOP, Software Engineering, DBMS/SQL, programming fundamentals, C#, and software-development practice; designed lectures, laboratories, assessments, rubrics, and feedback.
- Supervised final-year projects and contributed to curriculum development.

RESEARCH CONTRIBUTIONS

Efficient graph unlearning. Develops request-centered and subgraph-local update strategies that reduce the scope of retraining while preserving utility and privacy-oriented evaluation.

Transparent verification. Introduces explanation-shift and structural diagnostics for assessing whether GNN unlearning has removed the influence of designated graph entities.

Realistic and auditable benchmarks. Connects natural-language erasure requests and data-quality-triggered forget sets to concrete graph targets, with reproducible provenance and multi-view evaluation.

Evidence synthesis and open science. Curates systematic reviews of application-oriented GNNs and retrieval-augmented recommenders, alongside public Python implementations and benchmark artifacts.

SELECTED PUBLICATIONS & MANUSCRIPTS

Imran Ahsan, Brig B. Gupta, Jinsung Kim, and Mucheol Kim “NodeVanisher: Efficient and Scalable Data Unlearning in GNNs through Targeted Subgraph Refinement.” *IEEE Transactions on Consumer Electronics*, accepted, 2026. **Accepted**

Imran Ahsan, Hyunwook Yu, Jinsung Kim, and Mucheol Kim. “Forget and Explain: Transparent Verification of GNN Unlearning.” *Proceedings of the 19th ACM International Conference on Web Search and Data Mining (WSDM ’26)*, pp. 1063–1067, 2026. [PAPER](#) [CODE](#)

Imran Ahsan, Jinsung Kim, and Mucheol Kim. “From Random Forget-Sets to Realistic Natural-Language Deletion Requests.” *Proceedings of the ACM Web Conference 2026 (WWW ’26)*, pp. 8741–8744, 2026. [PAPER](#) [CODE](#)

Imran Ahsan, M. W. Anwar, J. Kim, and Mucheol Kim. “Unlocking the Power of Graph Neural Networks—A Systematic Literature Review of Application-Oriented GNN Studies.” *Computer Modeling in Engineering & Sciences*, 148(1), Article 3, 2026. [PAPER](#) [ARTIFACTS](#)

Minhyeok Choi, **Imran Ahsan**, Hyunwook Yu, Taeyoung Choe, and Mucheol Kim. “The Semantic Design Space of Retrieval-Augmented Recommender Systems: A Systematic Review of LLM-Based Approaches.” *Computers, Materials & Continua*, 88(1), Article 3, 2026. [PAPER](#)

[SUBMITTED](#) **Imran Ahsan** et al. “GDPRBench: A Controller-Side Benchmark for Accuracy-Triggered Graph Unlearning.” *Journal of Big Data*, manuscript submitted, 2026.

[SUBMITTED](#) **Imran Ahsan** et al. “Retrain-Aligned Explanation Auditing for GNN Unlearning.” *Expert Systems with Applications*, manuscript submitted, 2026.

Complete and continuously updated publication record: [GOOGLE SCHOLAR](#)

RESEARCH DIRECTION

Building reliable graph-based data-science systems that can explain, update, and remove learned information responsibly. Current priorities include scalable unlearning, human-understandable deletion verification, provenance-aware benchmarks, and integration of data-quality and governance signals into model-maintenance workflows.

TEACHING, CURRICULUM & SUPERVISION

University Teaching Portfolio

6 years

- Core courses: Object-Oriented Programming with Java; Software Engineering; Database Management Systems and SQL; Programming Fundamentals; C#/.NET; software-development practice.
- Teaching design: lecture–laboratory integration, project-based assessment, rubric development, formative and summative feedback, examination design, and marking.
- Student development: final-year project supervision, mentoring across varied programming backgrounds, and guidance in software design, database-backed applications, testing, implementation, and technical documentation.

PROFESSIONAL SOFTWARE ENGINEERING

Software Developer / Enterprise Systems Developer — Pakistan

4+ years

- Developed ASP.NET-based university portals, admissions systems, student-service modules, HR systems, and business applications.
- Technology stack: C#, ASP.NET, JavaScript, HTML/CSS, Microsoft SQL Server, MySQL, and Oracle.

TECHNICAL SKILLS

Graph AI & Trustworthy ML	Graph neural networks; graph and machine unlearning; explainable AI (XAI); privacy diagnostics; responsible model update; data-quality-aware evaluation.
Data Science & Research	Experimental design; benchmark construction; reproducible evaluation; systematic literature reviews; NLP and information retrieval; retrieval-augmented systems.
Programming & ML	Python; PyTorch; PyTorch Geometric; scikit-learn; Java; C#; ASP.NET; JavaScript.
Databases & Web	SQL; Microsoft SQL Server; MySQL; Oracle; database-backed enterprise applications; HTML/CSS.

HONORS & ACADEMIC HIGHLIGHTS

- Chung-Ang University Young Scientist Scholarship (CAYSS), 2023.
- BrainKorea (BK21) Scholarship Awarded. 2023

REFERENCES

Professor Mucheol Kim — Data Intelligence Laboratory, Chung-Ang University, Seoul, Republic of Korea.
Additional references are available on request.